

The use of online annotations in reading instruction and its impact on students' reading progress and processes

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Abstract

Studies suggest that the incorporation of online annotations in reading instruction can improve students' reading comprehension. However, little research has addressed how students use online annotations in their reading processes and how such use may lead to their improvement. This study thus adopted Reciprocal Teaching (RT) as an instructional framework to support students' reading comprehension progress and processes, facilitated by the use of online annotations. A total of 54 English language learners at a university were recruited to read with online annotations based on the RT procedure, namely predicting, clarifying, questioning, and summarizing. The data collected included the students' scores on pre- and post-tests and their participation records in a collaborative learning environment. The results reveal that the students enhanced their reading comprehension after the intervention. Their reading processes were also analyzed, and the major differences between the groups making more and less progress were identified. Those who made more progress not only frequently reviewed their previously generated predictions, clarifications, questions, and summaries but also actively provided feedback to their peers in a reciprocal manner. Pedagogical implications and recommendations are discussed.

Keywords: reading comprehension, online annotations, reciprocal teaching, reading processes

1 Introduction

The growth in online reading material has shifted readers' paper-based reading activities to web-based ones. The affordances of Web 2.0 technology, such as cross-platform connections, collaborative content creation, and (a)synchronous communication, have

further provided readers with opportunities for processing information through active interactions with others (Greenhow, Robelia & Hughes, 2009). As a means of enhancing online reading activities, many online annotation systems have been developed, such as Diigo (Kawase, Herder & Nejd, 2009), Annotea (Hwang, Wang & Sharples, 2007), and EDUCOSM (Nokelainen, Miettinen, Kurhila, Floréen & Tirri, 2005). These systems provide mechanisms that allow readers to annotate and discuss online texts synchronously or asynchronously, fostering new levels of knowledge by accumulating information from many users to facilitate deeper reading (Glover, Xu & Hardaker, 2007; Kawase *et al.*, 2009).

Although a number of studies on the use of online annotations in language classes have shown that such usage can improve comprehension, they tend to only report students' reading performance using quantitative approaches, and much less is revealed about the actual processes of how students use online annotations to enhance reading comprehension. To address this issue, the current study not only examined students' progress in reading comprehension, but also explored the processes of how they used online annotations in a collaborative learning environment. Specifically, the following two research questions were addressed:

1. Did the students make progress in their reading comprehension via online annotations?
2. What processes emerged to characterize how online annotations lead to students' reading comprehension?

2 Literature review

2.1 Problems of English as a foreign language (EFL) reading instruction in Taiwan

Reading and comprehending English academic texts is crucial for EFL students, as it enables them to obtain domain knowledge in their respective disciplines (Liu, Chen & Chang, 2010; Salinger, 2003). However, Shen and Huang (2007) noted that many college students in Taiwan do not have the level of English proficiency required to comprehend English textbooks. According to Yang (2010), up to 70% of EFL students are unable to read as effectively as they should. Moreover, when reading in a foreign language, few students are able to select effective strategies because they lack the related knowledge and metacognitive practices (Yang, 2010; Yeh & Yang, 2011). That is, if EFL students do not have the cognitive and metacognitive strategies needed to remedy their comprehension breakdowns, then they are likely to face many difficulties when reading English texts.

There are several factors that can account for the poor reading abilities among EFL students in Taiwan, including high student–teacher ratios, limited reading instruction, lack of engagement in reading comprehension processes, and insufficient reading strategy training. Firstly, the high student–teacher ratios prevent teachers from dealing with students' individual differences, and since the students thus receive little teacher feedback, they have few opportunities to resolve their comprehension problems (Yang, 2010). Secondly, the limited instructional time available for English classes does not provide enough opportunities for students to read in English (Lan, Sung & Chang, 2007). Thirdly, EFL students are often not trained to become active readers who are able to interact with the texts and monitor their comprehension. Many studies have also shown that students who

rarely engage in reading processes often have difficulty in comprehending texts (Dreyer & Nel, 2003; Nation & Norbury, 2005; Yeh & Yang, 2011). Finally, the majority of EFL reading instruction in Taiwan focuses mainly on grammar rules and vocabulary knowledge, rather than on teaching how to take an active role in reading processes (Liu *et al.*, 2010). Many studies have shown that students who use reading strategies have better reading comprehension than those who do not (Brantmeier, 2002; Singhal, 2001). Consequently, most college students in Taiwan may have difficulty in comprehending their English reading texts due to a lack of knowledge of reading strategies.

2.2 Online annotations in reading instruction

One possible solution to resolving the above-mentioned problems in EFL reading instruction is to incorporate the use of annotations. Annotations refer to “any additional content that is directly attached to a resource and that adds some implicit or explicit information in many different forms” (Nunes, Kawase, Dietze, de Campos & Nejd, 2012: 52), such as underlines, circles, highlights, glossaries, or explanatory notes. These different types of annotation have been proposed to facilitate EFL students’ reading, as they can help readers to: (a) learn new vocabulary, (b) reduce their cognitive load in reading, (c) understand the relations that exist among various pieces of textual information, and (d) better reflect on the textual information (e.g., AbuSeileek, 2011; Johnson, Archibald & Tenenbaum, 2010; Nunes *et al.*, 2012; Tseng, Yeh & Yang, 2015). In short, annotations can help students take appropriate actions to deepen their reading through constant monitoring of their comprehension (Kawase *et al.*, 2009).

In recent years a growing number of printed texts have been published electronically, and Web 2.0 technologies have made annotating online texts more feasible for readers. Scholars have thus indicated that online annotations can help second language (L2) learners read in various ways due to their characteristics of accessibility, availability, and shareability (Chen, Hwang & Wang, 2012; Glover *et al.*, 2007). Among these, the shareability function allows users to share their annotated texts or comments with other members in a group, who can then respond to each other’s work (Gao, 2013). This sharing and responding feature also encourages readers to discuss the focal text in a collaborative learning environment. Therefore, the use of online annotations can help learners to learn from their peers in a collaborative context through sharing experiences and constructing knowledge (Robert, 2009).

Several researchers have concluded that online annotations have a positive impact on reading comprehension (AbuSeileek, 2011; Johnson *et al.*, 2010; Tseng *et al.*, 2015). For example, AbuSeileek (2011) conducted an experiment to investigate the effects of reading EFL texts using different forms of online annotation in a hypermedia environment. A total of 78 undergraduate EFL learners were recruited to read and summarize seven texts and then take a vocabulary test. The post-tests results revealed that those who read the texts with online annotations outperformed those who did not, in terms of text memory and vocabulary acquisition. Johnson *et al.* (2010) conducted two experiments to investigate student learning outcomes with and without an online annotation system, SAM-LS. One of the experimental results showed that 267 college students who used the online annotation system improved their reading comprehension and metacognitive skills in both individual and group modes. They also indicated that the group mode improved students’ learning

outcomes better than in the individual mode when both used the online annotation system. Similarly, Tseng *et al.* (2015) reported the positive effects that an online annotation system, A.nnotate, had on reading comprehension. They probed into how different usages of online annotations led to higher levels of reading comprehension in EFL students, at surface-based, text-based, and situation-based levels respectively. The findings from the studies mentioned above all pointed to the positive impact of reading with online annotations.

While many studies have reported the positive effects of online annotations on reading comprehension, some gaps can be identified in the literature. First, the effect of online annotations on reading comprehension research was often evaluated through reading comprehension tests, ignoring the actual processes of how students used online annotations to facilitate their reading comprehension. Without this information, teachers cannot understand to what extent online annotations are useful in this context, and thus cannot provide suggestions about how to integrate online annotations to scaffold students' reading. In addition, the issue of careful instructional design in combination with online annotations still remains underexplored in tertiary EFL education. One such instructional design which may train students to strategically make the optimum use of online annotations is the framework of reciprocal teaching.

2.3 Related studies on reciprocal teaching (RT) and reading comprehension

RT is a teaching approach which encourages students to alternate "leading a dialogue centered on pertinent features of the text" (Palincsar & Brown, 1984: 117). Palincsar and Brown (1984) claim that RT can enhance students' reading comprehension significantly via the four strategies of predicting, clarifying, questioning, and summarizing. Firstly, predicting involves readers confirming or rejecting prior hypotheses based on the text. Secondly, readers are encouraged to clarify any confusion they encounter by using external resources and repeated reading. Next, questioning provides readers with the opportunity to make and negotiate meanings, by actively constructing question items for better comprehension. Finally, by summarizing the text, readers are engaged in distinguishing the important content from the less important parts, in order to integrate the former in their summaries (Doolittle, Hicks, Triplett, Nichols & Young, 2006).

Incorporation of RT into reading classes has been deemed an effective teaching approach with regard to various outcomes, such as improved reading comprehension (Fung, Wilkinson & Moore, 2003; Spörer, Brunstein & Kieschke, 2009; Yang, 2010), peer support and encouragement (Yang, 2010), metacognition enhancement (Fung *et al.*, 2003), and strategy acquisition (Spörer *et al.*, 2009). For example, Yang (2010) designed an online RT system to monitor the reading processes of 129 Taiwanese college students, and also investigated their reading comprehension. The system aimed to record students' reading processes via four functions supporting the four strategies of RT. The results showed that the online RT system not only helped the EFL learners improve their reading comprehension and gain peer support, but also helped the teacher better understand the students' learning difficulties to provide necessary scaffolding. In another study, Fung *et al.* (2003) used RT as an instructional design to investigate how English as a second language (ESL) 6th and 7th graders, who had recently migrated from Taiwan to New Zealand, comprehended an English expository text. The results indicated that RT improved not only the English reading competence of the Mandarin-speaking ESL students,

but also their metacognition. In a similar vein, Spörer *et al.* (2009) conducted an experimental study to compare RT strategy instruction with traditional instruction to measure the effects of the former on ESL elementary students' reading comprehension in German. The results revealed that the students who experienced RT in small groups and pairs gained higher reading scores and strategy acquisition in the post-test than those who were in the control group. In conclusion, with much research showing the positive influences of RT on learners with different school levels and learning backgrounds, this study was designed to understand if RT can serve as an appropriate instructional design to frame the incorporation of online annotations in a reading class.

3 Method

3.1 Participants

This study involved 54 non-English majors who registered in a freshman English course, *Vocabulary and Reading*, at a university in Taiwan. As EFL learners, they had limited English instruction, and the course under investigation was the only English-related course they took during the semester. Although the participants had been learning English for more than ten years, their English reading ability was regarded as low-intermediate based on the average score of 238 in the reading section of the Test of English for International Communication (TOEIC).

3.2 Research design

The reading class comprised weekly two-hour sessions over eighteen weeks. Before the reading instruction, the 54 participating students took the simulated TOEIC reading test as a pre-test in the first week. The students were also asked to annotate an assigned text with no instructional guidance. During the following two weeks, they were then introduced to the role of annotation in reading and how to effectively annotate a text using the RT procedure, involving the predicting, clarifying, questioning, and summarizing tasks. In the predicting task, the students had to add notes in the margins to write down their own predictions. In the clarifying task, they needed to mark and annotate unknown vocabulary with definitions, or derive the meanings of difficult concepts through group discussions. In the questioning task, they had to underline important sentences and add notes to generate possible comprehension questions for each other. In the summarizing task, they were asked to highlight key words and topic sentences, and then add notes in the margins to write down their summaries of the reading text.

In completing the four RT tasks, the students were divided into groups of three (eighteen groups in total), and asked to annotate six assigned articles (one per week) using Google Docs. In addition, they were encouraged to review their peers' work, ask for help, and engage in discussions with their peers or instructor in the online learning environment enabled by Google Docs. After completing annotations on each article, following the RT procedure, the students were required to share links to their work via a classroom management system on a weekly basis. At the end of the research intervention, all the students took another set of simulated TOEIC reading tests as the post-test to evaluate their reading progress.

3.3 Using Google Docs for online reading and annotations

Google Docs provided the collaborative learning context for this study through its various embedded functions, such as the annotation tool, document-sharing mechanism, and online chat rooms. Each function is briefly described below. The interface of Google Docs is available in many languages, such as English and Spanish, and users can select their preferred one.

The *annotation tool* is the main function the students use when undertaking the RT tasks. Students can annotate the title for prediction, annotate the unknown words or concepts with explanations for clarification, ask questions for peers to answer, and annotate the main ideas of each paragraph for summarizing. In this online setting, they can independently or collaboratively annotate the texts based on the RT procedure. The *document-sharing mechanism* allows students to share their notes, and invite others to collaboratively read and annotate the texts. In this online setting, students can observe their peers' work, have discussions with them, comment on their annotations, and respond to their comments. Students can use the *online chat room* function to clarify unknown words or concepts, and create different discussion threads with their peers. This function also allows the instructor to provide immediate feedback and assist the students if they encounter any reading difficulties.

3.4 Data collection and analysis

This study collected both quantitative and qualitative data, including the pre- and post-tests, and the students' participation records from Google Docs. With regard to the quantitative data, the students' scores in the pre- and post-tests were used to investigate any changes in their reading comprehension after using online annotations during the RT. Paired-samples *t*-tests were adopted to examine the students' progress between the pre- and post-tests, and the scores were used to divide them into two groups: (1) the "more progress" (MP) group of fourteen students with gain scores in the top 25% of the class; (2) the "less progress" (LP) group of thirteen students with gain scores in the bottom 25% of the class. Both the MP and LP groups' reading performances and processes were closely analyzed to identify any similarities and differences between them. Furthermore, one representative case was chosen from each group, using a typical case sampling technique, to delineate the salient features of their reading processes. As Bamberger, Rugh and Mabry (2011) stated, typical case sampling is a type of purposeful sampling in which, "subjects are selected who are likely to behave as most of their counterparts would" (p. 358). Accordingly, the two typical cases in this study (S1 from the MP group and S2 from the LP group) were purposefully selected based on the following criteria: (1) their pre-test scores were close to the class average, and (2) their gain scores from the pre- to the post-tests were close to the group average.

The qualitative data consisted of students' participation records from Google Docs, which were used to examine the processes of how they used annotations to support their reading comprehension. This study adopted Patton's (2002) four procedures of content analysis, namely coding, categorization, description, and interpretation, to analyze the students' reading processes with online annotations. Based on the participation records in Google Docs, the research team first coded meaningful statements from the participants by

reading all the participation records (coding). The meaningful statements were grouped into categories in terms of the students’ use of online annotations during the RT in the predicting, clarifying, questioning, and summarizing tasks (categorization). The researchers further described the students’ reading progress and processes with examples and illustrations (description). Finally, the researchers interpreted the students’ reading processes by providing explanations, making inferences, and drawing conclusions regarding how they used online annotations during the RT in each task (interpretation). Any disagreements about the coding that arose among the research team were resolved by discussion. The mean inter-rater reliability was .87, which was calculated based on percent agreement of all the participation records across different data sets.

4 Results

4.1 Students’ progress in their reading comprehension after using online annotations

The learning outcomes in the pre- and post-tests are presented in Table 1. Most of the 54 participants made satisfactory progress, with a significant difference ($t = 6.799, p < .001$) between the pre-test ($M = 238, SD = 44.481$) and post-test ($M = 276, SD = 51.340$). The effect size for Cohen’s d was .79, which indicated small to medium practical significance according to Plonsky and Oswald’s (2014) empirically-derived benchmarks for second language acquisition. Paired-samples t -tests were conducted to examine the progress made by the MP and LP groups. As shown in Table 2, a significant difference ($t = 13.491, p < .001$) between the pre-test ($M = 213, SD = 45.84$) and post-test ($M = 303, SD = 48.08$) was found for the MP group, with a very large effect size of 1.92. Moreover, there was also a significant difference ($t = 8.02, p < .001$) between the pre-test ($M = 248, SD = 34.16$) and post-test ($M = 262, SD = 33.26$) for the LP group, though with a

Table 1 Statistical results of the pre- and post-tests for all the students

	Min	Max	M	SD	<i>t</i>	Sig. (2-tailed)
Pre-test	139	287	238	44.48	6.80	.000*
Post-test	203	376	276	51.34		

Table 2 Statistical results of the pre- and post-tests for the MP and LP groups

Group	Min	Max	Mean	SD	<i>t</i>	Sig. (2-tailed)
MP Group (N = 14)						
Pre-test	139	287	213	45.84	13.49	.000
Post-test	203	376	303	48.08		
LP Group (N = 13)						
Pre-test	173	193	248	34.16	8.02	.000
Post-test	287	307	262	33.26		

negligible effect size of .42. These results indicated that both groups significantly improved their reading comprehension after using online annotations while reading, though the analysis demonstrated much higher statistical power for the MP group. Since both groups had similar levels of proficiency before the intervention, the difference in their learning outcomes can be attributed to the use of online annotations in their reading processes.

4.2 Representative cases: Reading processes with online annotations

In this study, one student from each of the MP and LP groups was chosen as a case study based on typical case sampling in order to gain more in-depth information about the reading processes involved. These students were labelled S1 and S2, respectively, in the following discussion. As shown in Figure 1, S1 took more actions in each RT task than S2: 34 more actions in predicting, 81 more actions in clarifying, 35 more actions in questioning, and 38 more actions in summarizing.

Overall, S1 and S2 underwent similar sequences in task completion during the RT, from predicting, clarifying, and questioning to summarizing. However, the key difference between S1 and S2 lies in the reviewing actions. As shown in Figure 2, when S1 completed the summarizing tasks, it was found that she often undertook recursive actions to revisit previous tasks and further revised previously generated predictions, clarifications, questions, and summaries. In contrast, S2 demonstrated less reviewing actions in task completion of RT.

Table 3 presents the frequency difference between the two students' use of online annotation actions in the four RT tasks. In the predicting task, S1 annotated the titles and highlighted key words or sentences more frequently to make initial predictions than S2 did. S1 also took many more actions to revise her previously self-constructed predictions. In the clarifying task, S1 undertook more annotation actions in terms of vocabulary items, complex sentences, confusing ideas, and interesting parts with personal reactions to interact and clarify with her peers. In the questioning task, S1 generated more comprehension

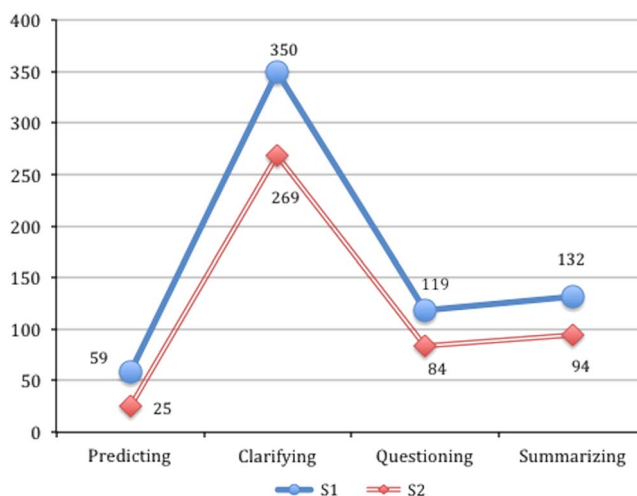


Fig. 1. S1's and S2's frequency of actions in each RT task

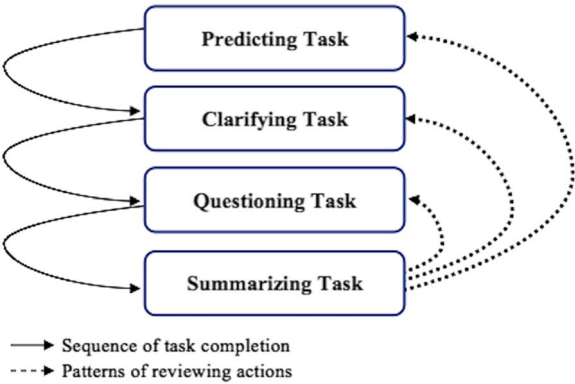


Fig. 2. S1’s task completion sequence and reviewing actions in RT

Table 3 S1’s and S2’s frequency of actions in RT

RT Tasks	Frequency	
	S1	S2
Predicting Task		
Annotate titles	7	5
Make initial predictions by highlighting keywords or sentences	28	11
Revise self-constructed prediction	24	9
Clarifying Task		
Annotate unknown words with definitions	144	135
Annotate complex sentences with grammatical usage	86	72
Annotate unknown concepts or confusing ideas	38	19
Annotate interesting parts with personal reactions	22	8
Provide suggestions for peers’ clarifications	32	22
Revise self-constructed clarifications	28	13
Questioning Task		
Annotate main ideas to generate comprehension questions	46	31
Annotate keywords to answer the generated questions	21	14
Provide suggestions for peers’ questions	33	28
Revise self-constructed questions	19	11
Summarizing Task		
Annotate keywords or main ideas	42	28
Write paragraph summaries	32	27
Provide suggestions for peers’ summaries	26	20
Revise self-constructed summaries	32	19

questions for peer interactions, and in turn annotated more key words and main ideas in response to her generated questions. In the summarizing task, S1 exerted more effort in annotating the key words or main ideas in order to compose paragraph summaries, and further revised her previous versions of summary. In sum, comparing the frequency of actions between S1 and S2, the results show that S1 was more actively engaged in the online reading processes with online annotations than S2.

The participation records from Google Docs enable further analysis of the processes of how S1 utilized online annotations to enhance her reading comprehension in these four RT tasks. In the predicting task, S1 first read the title “Restaurant Z” and then made a prediction about the text: “It might talk about how technology prevents humans from starving” (Figure 3). After reading the first paragraph, she went back on her previous prediction, stating: “I was wrong!” In making predictions, S1 marked the key words to monitor her own reading process with her own predictions. She then clarified her understanding of word meanings and discussed her prediction with peers.

In the clarifying task, S1 used the online chat room to answer her peers’ questions by clarifying the meaning of “digital dining” (Figure 4). To help her peers gain more information, she responded to their inquiries and then posted a link to the photo and further explained the new concept. In addition, S1 frequently used the online chat room to clarify confusing words, phrases, or concepts with her peers and thus improve her reading comprehension, as well as to provide additional external sources for them. She further provided her personal concerns over the issue. Next, she clarified her puzzle over the meaning of the word *tip* with the instructor and peer. From their responses she learned the meaning of the sentence, and became aware that the word *tip* has different meanings among the peer responses. With the ongoing discussions, S1 constantly used the online chat room to clarify those comprehension questions and also obtained feedback from her peers and the instructor.

In the questioning task, after S1 had read the fourth paragraph she highlighted the extract: “Now you may want to know how this change will affect you. Maybe you are afraid of losing your job. Let me assure you, you won’t.” Then she generated a question: “Will this new technology make people lose their jobs?” for her peers to answer, and provided a specific answer to double check whether her own response was consistent with those of her peers. Figure 5 illustrates this process. During the process of question generation and answer explanation, S1 not only highlighted the possible sentences for generating a question, but also went over the previously annotated information to propose a correct answer. By comparing her peers’ responses with her own, S1 could further evaluate her understanding of the reading text.

In the summarizing task, S1’s own annotations served as clues when she wrote her summary (Figure 6). After posting her summary, S1 received peer feedback both on local errors (e.g., plural nouns or verb agreements) and on global suggestions (e.g., coherence).

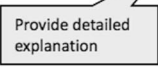
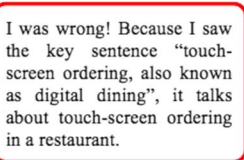
Prediction	Actions undertaken in reading with online annotations	Confirm or reject the previous prediction
It might talk about how technology prevents humans from starving. Maybe it would mention about the food shortage around the world.	Annotate the title “Restaurant Z.” Annotate the sentence “Next month, we will make a change so that we can really stand out: touch-screen ordering, also known as digital dining.” 	

Fig. 3. Summary illustration of S1’s predicting task recorded in Google Docs

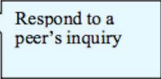
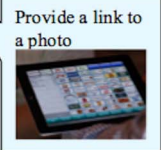
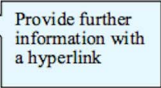
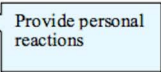
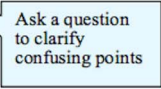
Participants		Interactions among S1, her peers (S3, S4), and the teacher	
S3:	Do you know what “digital dining” means?		
S4:	Maybe it is something related to “dinner.”		
S1:	I think it is a machine that can automatically allow people to order.		
	Here is the link to the picture. http://www.yelp.com/biz_photos/integrated-restaurant-systems-minneapolis-2?select=UTqF0qMwq1wpoOioV6YVTVw		
Teacher:	Good! Really helpful! How did you know that?		
S1:	I saw the relevant news story several months ago. http://www.mds.com.tw/product/product_o4.html		
S4:	Cool! It's amazing!		
S3:	It is convenient! Do we have this technology in Taiwan?		
S1:	I am not sure. Probably not yet. I was concerned that if we use this technology in the future, what else can a server do?		
Teacher:	Good point!		
S4:	Some people may lose their jobs, is that right? (worried)		
S1:	I don't understand this sentence “This exciting new technology will improve our customers' dining experience and increase your tips!” Why does the article say that “it can increase your tips”? “Tip” means “clue”, doesn't it? It doesn't make sense.		
Teacher:	Right! The word “tip” can be a verb or noun. How about checking the online dictionary and identifying which meaning fits in this context.		
S3:	“Tip” can also mean “service fee.”		
S1:	Good! This definition would fit in this context. Thank you ☺		

Fig. 4. 1. Summary illustration of S1’s clarifying task recorded in Google Docs

S1 then reviewed her previous annotations and took her peers’ suggestions to revise her first summary draft by correcting grammatical errors and adding the key concept (e.g., a digital dining machine) and transitional words like *thus* and *instead* into the final draft of her summary. In short, S1 not only frequently retrieved previous annotated sentences to revise her summary, but also benefited from her peers’ suggestions, which helped her pinpoint problems in her writing. By revising the summary, S1 obtained another chance to monitor and evaluate her understanding of the reading text through a process of repeated reviewing actions.

Although S1 and S2 shared similar tendencies when undertaking the four RT tasks with online annotations, there were still notable differences. In the predicting task, when S2

Sentence highlighted	
“Now you may want to know how this change will affect you. Maybe you are afraid of losing your job. Let me assure you: you won’t.”	
Participants	Interactions
S1:	<i>(Comprehension Question)</i> Will this new technology make people lose their jobs?
S3:	<i>(Peer Response)</i> I think they won’t because technology is just a tool.
S4:	<i>(Peer Response)</i> I don’t think so. According to this article, it stated that they wouldn’t lose their jobs. High-tech makes ordering more efficient, so that waiters can serve more customers.
S1:	<i>(Answer)</i> Both of your answers are correct. People won’t lose their job because this high-tech not only looks cool, but also decreases the number of tasks you have to do. Therefore, you have time and energy to serve more tables without feeling stressed!

Fig. 5. Summary illustration of S1’s questioning task recorded in Google Docs

rejected the previous predictions, he seldom further revised his previous incorrect prediction like S1 did. He would simply note: “I made a wrong prediction!” without providing any further explanation. Similarly, in the clarifying task, S2 seldom annotated unknown words, sentences, concepts, or confusing ideas with peers about the text as S1 did, but instead paid more attention to clarifying word meanings and grammar rules. For example, when reading the sentence: “All you have to do is bring the food to the table”, S2 used the online chat room to ask his peers: “Is this sentence grammatically correct? I think the verb should be *are*.” The above example shows that S2 mainly used this task to deal with the text at the micro-level. S2 also took fewer actions to annotate interesting parts with personal reactions in the text than S1 did. In the questioning task, after generating questions S2 did not annotate key words or main ideas to answer them, but instead only provided his peers with simple answers, which failed to convince them. For instance, he generated the comprehension question: “Why is Restaurant Z so popular?” After his answer “Because they have a new high-tech” was posted, one of his peers replied: “The real reason is that they have a digital dining machine.” That is, since S2 marked fewer key words and main ideas for answers, he was not able to answer his self-constructed comprehension questions as precisely as S1. With regard to the summarizing task, the results showed that S2 exerted less effort in reviewing his previous annotated key words and main ideas for summary writing than S1 did. After receiving peer feedback, S2 simply corrected local errors even though his peers pointed out some logical problems in his summary.

With regard to the teacher–student and student–student interactions within the collaborative learning environment of Google Docs, S1 and S2 both undertook the

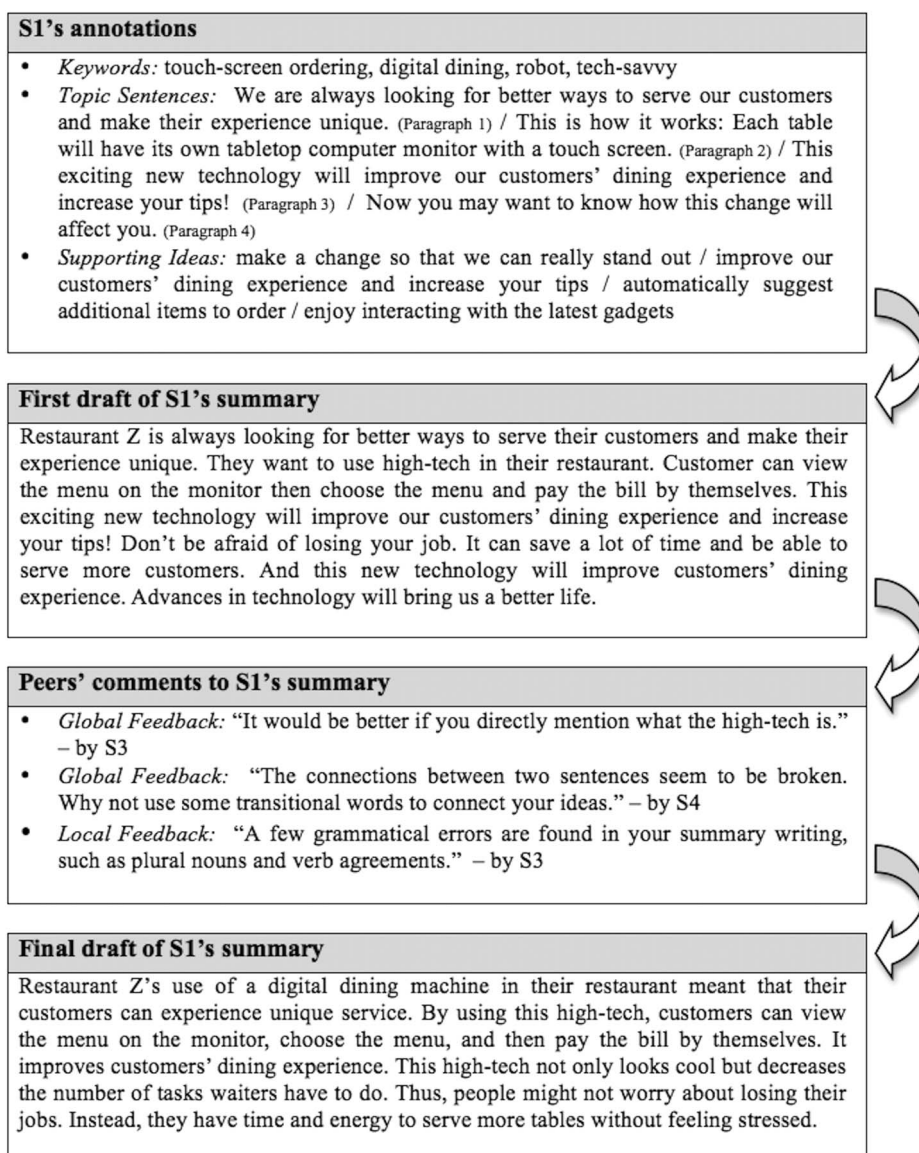


Fig. 6. Summary illustration of S1's summarizing task recorded in Google Docs

four RT tasks with online annotations collaboratively with the others in their groups, and the instructor could provide appropriate assistance if needed. S1 and S2 were thus able to improve their reading comprehension by giving/receiving feedback to/from peers and based on the instructor's adaptive scaffolding while completing the tasks given to them. The instructor also served as a facilitator to make sure that the students undertook the tasks properly, helped them solve any reading difficulties that arose, and provided scaffolding so that they could attain a higher level of reading comprehension.

5 Discussion and conclusion

This study illustrated how the instructional design of RT (consisting of predicting, clarifying, questioning, and summarizing tasks) was integrated into a reading class with the use of online annotations, with the overall aim of enhancing the students' reading comprehension and reading processes. The results of the pre- and post-tests indicated that all of the students significantly improved their reading comprehension with the support of online annotations. This finding is consistent with previous studies (e.g., AbuSeileek, 2011; Johnson *et al.*, 2010) which suggest that online annotations have positive effects on reading comprehension. Some significant differences were found in the reading processes of the more and less progress groups. The MP group performed many more actions in each RT task when annotating reading texts. Additionally, they revisited previously generated predictions, clarifications, and questions while completing summaries compared to the LP group. As a result, the use of these RT tasks, coupled with the use of online annotations, was able to enhance the students' reading comprehension to different extents.

In order to delve into these EFL students' reading processes through online annotations, S1 and S2 were selected from the MP and LP groups, respectively, to gain insights into how the two groups carried out online reading in the four RT tasks. In the predicting task, although both students activated their prior knowledge to make predictions before reading, S1 made more effort in verifying her predictions by repeatedly comparing them with the texts and providing more details to correct her wrong predictions after reading. As Duke and Pearson (2002) note, good readers are those who "draw from, compare, and integrate their prior knowledge with material in the text" (p. 206). Fielding, Anderson and Pearson (1990) further suggest that the process of explicit comparison and verification after prediction is a determining factor that facilitates participants' understanding of a text. That is, better reading comprehension can be achieved when students make frequent comparisons between their existing knowledge and the text, and then assess their previous predictions based on this.

In the clarifying task, S1 demonstrated greater involvement in annotating words, sentences, and confusing concepts as well as providing personal reactions than S2, who merely attempted to understand unknown words. This corresponds to findings from Tseng *et al.* (2015), in which deep readers among EFL students tended to mark the main ideas and add notes to reflect on the reading texts using in-depth analysis and interpretations. Furthermore, S1 had a higher frequency of annotating interesting parts with personal reactions and responding to peers' inquiries than S2 did. By clarifying the information that causes comprehension problems, students can not only receive peer support but also answer each other's inquiries, and thus students can monitor and improve their own reading comprehension (Yang, 2010).

In the questioning task, S1 showed greater engagement in generating questions and responding to questions from peers. This suggests that the level of engagement in producing and responding to questions can trigger deeper reading comprehension. In this task, the students needed to fully understand the text before they were able to produce comprehension questions in the online chat room with their peers. In order to answer their peers' questions, they had to read the text repeatedly, search for information related to the question, and then engage in deep thinking about the information in the text (Yang, 2010). Moreover, disagreements on peers' questions or answers could also prompt students to go

over the text to re-evaluate their understanding (Yeh & Yang, 2011). By actively engaging in the process of question generation, students can monitor their own comprehension of the texts with their peers, thus enhancing their reading comprehension (Spire & Donley, 1998).

In the summarizing task, S1 was found to be more active in revising her work after receiving peer feedback and undertook more reviewing actions than S2. This suggests that a greater frequency of reviewing the summaries offered more chances for students to absorb what they read, thus enhancing their reading comprehension. As claimed by some researchers (e.g., Brown, Day & Jones, 1983; Duke & Pearson, 2002), students who frequently participate in constructing and revising summaries can improve their reading by monitoring and verifying their comprehension of the focal texts during the processes of summary writing. Taken together, the reviewing action in the four RT tasks was a pivotal factor contributing to students' improved reading comprehension when undertaking online annotations. This result confirms previous studies, in that when students consistently monitor, evaluate, and self-regulate during the reading processes, their comprehension is likely to be enhanced (e.g., Dreyer & Nel, 2003; Spörer *et al.*, 2009).

As previous studies have noted, computer-supported collaborative learning maximizes learning processes and performance, since these technologies can help students to actively engage with peers whose opinions they deem valuable. The affordances of Google Docs used in this study enabled the students to read each other's thoughts in the form of annotations, and give each other feedback without the usual classroom constraints of time and space. That is, students could learn the reading texts and explore new knowledge collaboratively in the collaborative learning environment of Google Docs. The participation records related to the online annotations allowed the students to monitor and compare their reading processes with those of their peers, which in turn helped them become more aware of their own misunderstanding of the texts. Moreover, the use of the online chat room also served as a collaborative learning forum where students could generate discussion threads for social constructive feedback, thus exchanging their learning experiences, clarifying confusing parts, and generating solutions to any problems that arose. In addition, the sharing mechanism of Google Docs permitted the teacher to provide students with appropriate scaffolding by identifying and solving reading problems through discussion. If any ineffective use is found, then the teacher should clarify or modify the instructional approach to make the learning more effective (Duke & Pearson, 2002). Overall, this study supports the idea that providing students with opportunities to use online annotations in a collaborative learning environment can help them engage in reading and improve their comprehension with reciprocal efforts.

This study has several implications for reading instruction and research. Firstly, since students' reading comprehension significantly improved after reading instruction with online annotations, language teachers might be encouraged to apply an appropriate instructional design such as RT in this study, to guide and scaffold students' use of online annotations in their reading processes. With the caveat in this study that technology might not be the sole variable accounting for the difference in learning, the use of online annotations should be coupled with carefully designed reading instruction. Secondly, to help students gain a better understanding of the text, teachers should encourage them to annotate the text collaboratively not only at word level, but also at higher-order levels. In addition, teaching students to provide more effective comments is important, since constructive feedback can also help them to achieve better reading comprehension through

the process of constructing and reviewing their summaries. Thirdly, it is suggested that language teachers prompt students to play more active roles in revising their previously generated predictions, clarifications, questions, and summaries, since taking reviewing action appears to play a pivotal role in augmenting reading comprehension. This can also be done even in a large class, with an appropriate instructional design that engages students in a collaborative learning environment. To conclude, the overall results of this study have shown that the reciprocal nature of collaborative dialogues that occurs through online annotations serves to support students' reading process, which in turn leads to their improved comprehension. Future studies should thus build on this understanding when implementing reading instruction with online annotations.

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